

Data Logging System for Physiological Parameters

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Introduction

- ▶ Monitoring physiological parameters is an important part of modern healthcare systems.
- ▶ Continuous monitoring helps doctors detect abnormal health conditions at an early stage.
- ▶ Physiological parameters such as Heart Rate, Blood Pressure, Oxygen Saturation and Temperature indicate the health status of a patient.
- ▶ Traditional monitoring systems require manual observation by medical staff.
- ▶ Automated monitoring systems can continuously track patient data and improve response time.
- ▶ Micro-controller based systems allow efficient collection and processing of physiological data.
- ▶ Data logging systems help store and analyze patient health information over time.
- ▶ Alert systems can notify medical staff when abnormal readings occur.



Problem Statement

- ▶ Continuous monitoring of patient physiological parameters is critical in healthcare environments.
- ▶ In many cases, patient parameters are monitored manually, which can be time consuming and prone to human error.
- ▶ Sudden abnormal changes in physiological parameters may not be detected immediately.
- ▶ Lack of automated alert systems can delay medical response during critical conditions.
- ▶ There is a need for a system that can simulate, monitor and analyze physiological data continuously.
- ▶ The system should also generate alerts when abnormal parameter values are detected.



Objectives

- ▶ To simulate physiological parameter data using the TM4C123GXL microcontroller.
- ▶ To collect and use the simulated data as input for the monitoring system.
- ▶ To analyze and visualize physiological parameters using graphical representation.
- ▶ To monitor parameters such as Heart Rate, SpO2, Respiratory Rate, Temperature and Blood Pressure.
- ▶ To detect abnormal parameter values based on predefined normal ranges.
- ▶ To implement an alert generation system for abnormal readings.
- ▶ To broadcast alerts to systems running the receiver program.



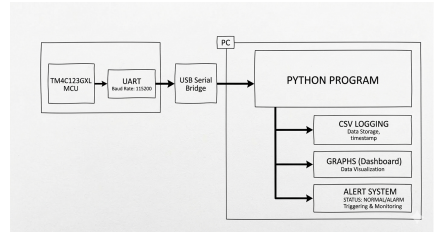
System Overview

- ▶ The system simulates physiological parameter data using the TM4C123GXL microcontroller.
- ▶ The generated data represents patient health parameters such as Heart Rate, SpO2, Respiratory Rate, Temperature and Blood Pressure.
- ▶ This simulated data is transmitted to a monitoring program.
- ▶ The monitoring system processes the incoming data for analysis.
- ▶ Graphs are plotted to visualize trends of each physiological parameter.
- ▶ The system checks whether the values fall within normal medical ranges.
- ▶ If abnormal values are detected, an alert message is generated.
- ▶ The alert is broadcasted to all systems running the receiver program.



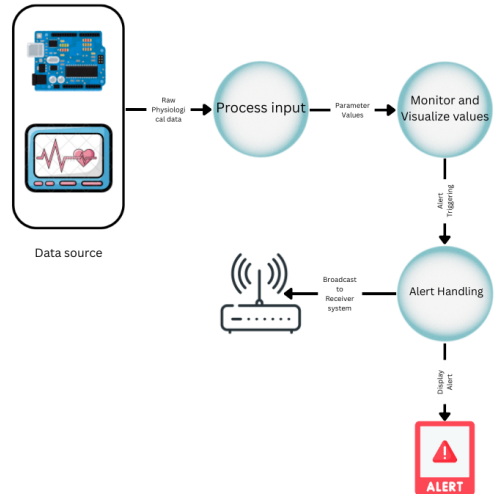
System Architecture

- ▶ TM4C123GXL microcontroller generates simulated physiological parameters.
- ▶ Data is transmitted using UART serial communication.
- ▶ The UART data is received by the computer through the USB serial interface.
- ▶ A Python monitoring program reads the incoming data through the COM port using PySerial.
- ▶ The received data is stored in a CSV file.
- ▶ Matplotlib generates graphs for visualization of physiological parameters.
- ▶ The system detects abnormal values and generates alerts.



System Workflow

- ▶ Data simulation using TM4C123GXL
- ▶ Transmission of physiological data
- ▶ Data reception by monitoring system
- ▶ Data processing and monitoring
- ▶ Graphical visualization of parameters
- ▶ Detection of abnormal values
- ▶ Alert generation
- ▶ Broadcast alerts to receiver systems



Software Tools Used

- ▶ **Code Composer Studio (CCS)** Used to write, compile and upload embedded C programs to the TM4C123GXL microcontroller.
- ▶ **Python** Used to receive and process the physiological data sent by the microcontroller.
- ▶ **Pandas** Python library used for storing and handling physiological data in structured format.
- ▶ **Matplotlib** Used to plot graphs for visualization of physiological parameters over time.
- ▶ **PySerial** Python library used for serial communication between TM4C123GXL and the monitoring system.



Implementation Methodology

- ▶ TM4C123GXL microcontroller is programmed using Code Composer Studio.
- ▶ Random number generation is used to simulate physiological parameters such as Heart Rate, SpO2, Respiratory Rate, Temperature and Blood Pressure.
- ▶ The generated data is transmitted to the computer using UART serial communication.
- ▶ A Python monitoring program receives the data through the USB serial port.
- ▶ The received data is stored in a CSV file for data logging.
- ▶ Graphs are generated using Matplotlib to visualize physiological parameters.
- ▶ The system checks parameter values and generates alerts when abnormal readings are detected.



Physiological Parameters Monitored

- ▶ Heart Rate (HR)
- ▶ Oxygen Saturation (SpO2)
- ▶ Respiratory Rate (RR)
- ▶ Body Temperature
- ▶ Blood Pressure (BP)

	Timestamp	PatientID	HR	BP	SpO2	Temp	RR
1	20260311192813	P12345	118	171/106	83	38.5	27
2	20260311192816	P12345	94	110/80	99	36.4	12
3	20260311192819	P12345	75	117/83	99	36.7	16
4	20260311192822	P12345	86	123/75	96	36.9	12
5	20260311192825	P12345	92	116/74	98	36.7	15
6	20260311192828	P12345	86	110/74	100	37.0	16
7	20260311192831	P12345	65	170/69	82	36.6	21
8	20260311192834	P12345	81	124/73	99	36.9	16
9	20260311192837	P12345	84	114/74	96	36.6	15
10	20260311192840	P12345	78	121/74	96	36.4	13
11	20260311192843	P12345	67	122/77	100	36.7	12
12	20260311192846	P12345	82	112/83	100	36.7	14
13	20260311192849	P12345	69	112/83	99	37.0	18
14	20260311192852	P12345	86	111/78	96	36.5	16
15	20260311192855	P12345	65	114/80	99	36.8	13
16	20260311192858	P12345	70	125/76	97	36.4	14
17	20260311192901	P12345	59	111/80	91	39.0	23
18	20260311192904	P12345	72	124/84	99	36.7	14
19	20260311192907	P12345	83	112/78	98	37.1	12



Physiological Data Simulation

- ▶ The TM4C123GXL micro-controller is used to generate simulated physiological data.
- ▶ Random number generation techniques are used to create realistic parameter values.
- ▶ The generated data mimics real patient physiological signals.
- ▶ Each parameter is generated within medically accepted ranges.
- ▶ The simulated data is transmitted to the monitoring system for further processing.
- ▶ This approach allows testing of the system without requiring actual medical devices.



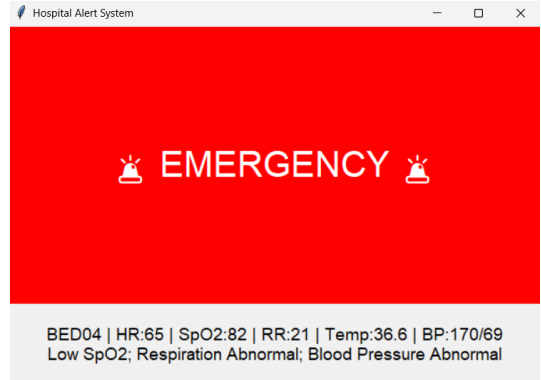
Graph Plotting

- ▶ Physiological data is processed and stored.
- ▶ Graphs are plotted for each physiological parameter.
- ▶ Visualization helps understand parameter trends over time.
- ▶ Real-time graph updates allow continuous monitoring of patient health.
- ▶ Graphs help identify sudden changes or abnormal patterns in the data.
- ▶ Abnormal conditions can be easily detected through visual analysis.



Alert Generation System

- ▶ The system continuously monitors incoming data.
- ▶ Each parameter is compared with predefined normal ranges.
- ▶ If abnormal values are detected, alerts are generated.
- ▶ Alerts are broad-casted to all systems running the receiver program.



Work Completed in Phase 1

- ▶ Simulated physiological data using TM4C123GXL micro-controller.
- ▶ Used simulated data as input to the monitoring system.
- ▶ Plotted graphs for individual parameters.
- ▶ Implemented alert detection system.
- ▶ Broadcasted alerts to receiver systems.



- ▶ Integrate real physiological sensors with micro-controller.
- ▶ Implement real-time physiological data logging.
- ▶ Develop advanced centralized monitoring dashboard.
- ▶ Implement the connection with USB, Bluetooth and WiFi
- ▶ Improve alert notification mechanisms.



Conclusion

- ▶ Successfully simulated physiological parameters using the TM4C123GXL microcontroller.
- ▶ Transmitted the generated data to the monitoring system using UART serial communication.
- ▶ Implemented data logging and graphical visualization of physiological parameters.
- ▶ Developed a basic alert detection system for abnormal parameter values.
- ▶ Demonstrated the feasibility of a microcontroller-based physiological data logging system.



References I

-  Texas Instruments. TM4C123GXL LaunchPad Microcontroller Datasheet
-  Texas Instruments. TivaWare Peripheral Driver Library User Guide
-  Python Software Foundation. PySerial Documentation
-  Chokri Baccouch and Chaima Bahar. Embedded system for ecg monitoring and fatigue detection using machine learning. IJACSA, 16(9), 2025. Embedded system for ecg monitoring
-  Guang-Zhong Yang et al. Real-time wearable physiological monitoring system for homehealthcare. Sensors, 22(1), 2022 physiological monitoring system for homehealthcare.



Many Thanks

